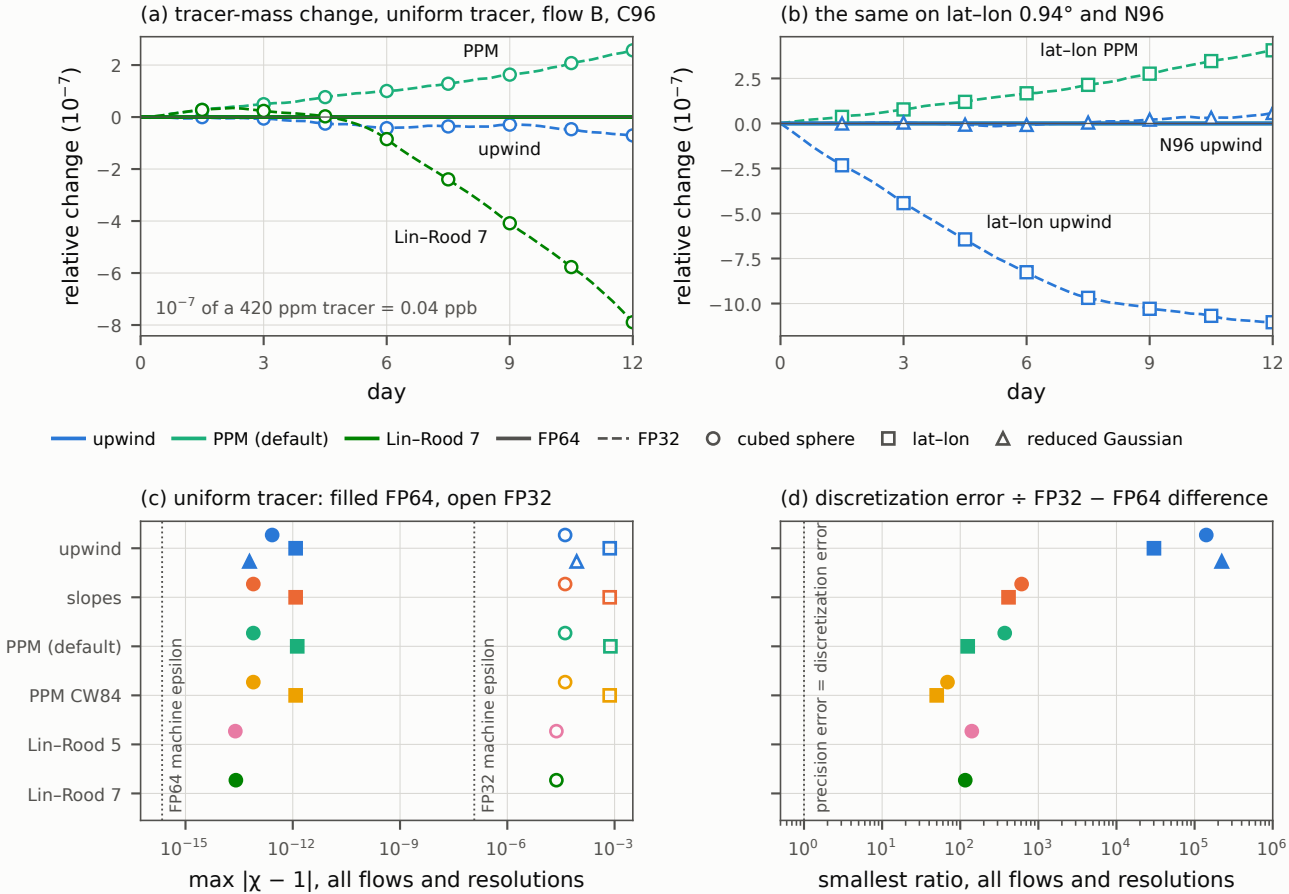
(a) tracer-mass change, uniform tracer, flow B, C96
relative change (10^{-7})
PPM
upwind
Lin–Rood 7
10^{-7} of a 420 ppm tracer = 0.04 ppb
day
(b) the same on lat–lon 0.94° and N96
lat–lon PPM
N96 upwind
lat–lon upwind
upwind
PPM (default)
Lin–Rood 7
FP64
FP32
cubed sphere
lat–lon
reduced Gaussian
(c) uniform tracer: filled FP64, open FP32
upwind
slopes
PPM (default)
PPM CW84
Lin–Rood 5
Lin–Rood 7
FP64 machine epsilon
FP32 machine epsilon
max |χ − 1|, all flows and resolutions
(d) discretization error ÷ FP32 − FP64 difference
precision error = discretization error
smallest ratio, all flows and resolutions